

تبارك الأكاديمية السعودية الرقمية لـ

Jadel Bader Aldaghmi

جادل بدر الدغمي

On successfully completing **Game Development Bootcamp** in a journey that lasted **13 weeks** From **24 August 2025** to **20 November 2025**

Issue date **20 November 2025**

إتمام معسكر همّة لتطوير الألعاب بنجاح بعد رحلة استمرت لمدة **13 أسبوع** في الفترة من **1 ربيع الأول 1447** الى **29 جمادى الأول 1447**

تم إصدارها بتاريخ **29 جمادى الأول 1447**

WISHING YOU A CONTINUOUS SUCCESS

مع تمنياتنا بدوام التوفيق والنجاح

Amal A. Aljuhaiman

الرئيس التنفيذي للأكاديمية السعودية الرقمية

أمل عبدالكريم الجهيمان



Bootcamp Syllabus

WEEK 1

Orientation

- Setup and installation of required software.

WEEK 2

GameKit

- Introduction to 3D Space Modeling Projects & Pipeline
- Introduction to GameKit sample games
- The game design process
- AI Applications in Game Development

WEEK 3

Basic Game Design

- Modeling Tools & Techniques Modeling Workflows
- Rules and game mechanics
- Actors and components in Blueprint

WEEK 4

Modelling

- Modeling for Realtime 3D Principles of UV Mapping
- Game systems
- How systems relate to features

WEEK 5

Unreal

- Intro to Unreal and Whiteboxing Modeling in Unreal and File Management
- The illusions of games
- Creating a game experience
- Drafting a game concept

WEEK 6

Niagara

- Basics with Materials
- Intro to Niagara
- Iteration as part of the design process
- Defining fun
- The pillars of a game concept
- Prototypes and how they're used

WEEK 7

Niagara

- Exporting and Importing Meshes and Animations
- Intro to Sequencer Animation Landscapes
- Communication in game design
- Understanding your audience
- Puzzles and communicating with players

WEEK 8

Rigging

- Animating with Control Rig Metahuman and Interactive Characters
- Documentation in game design
- Types of documentation
- Understanding your audience
- Writing a pitch

Bootcamp Syllabus

WEEK 9

Rig

- Facial Performance Capture
- Introduction to Rigging
- Themes in game design
- Worldbuilding

WEEK 10

Unreal Rig

- Character Rigging Basics
- Unreal Control Rig
- Sound in games
- How to implement basic audio effects
- User interfaces

WEEK 11

Landscapes

- Creating a Landscape
- Landscape Materials & Water Systems
- Visuals in games
- Lighting
- Animation

WEEK 12

Lighting Techniques

- Foliage And Debris
- Lighting Systems in Unreal
- Testing in games
- Troubleshooting and bug hunting

WEEK 13

Level design & final project

- Levels and World Partition
- Planning & Layout
- Polishing the final project